case study 1

(1. Phase: Ask)

1. Scenario & Background

• The Company: Cyclistic is a bike-share company featuring a fleet of over 5,800 geotracked bicycles and 692 docking stations across Chicago.

• Customer Segments: Cyclistic offers flexible pricing options, including single-ride passes and full-day passes (categorized as casual riders), alongside annual subscriptions (categorized as member riders).

• Financial Insight: Finance analysts have concluded that annual members are much more profitable than casual riders.

• The Stakeholders: Lily Moreno (Director of Marketing) believes the company's future growth depends on maximizing the number of annual memberships rather than focusing on attracting all-new customers.

2. Business Task

• Core Objective: Analyze 12 months of historical Cyclistic trip data to understand how annual members and casual riders use the service differently.

• Primary Goal: Design data-driven marketing strategies and actionable recommendations to convert casual riders into annual members, pending approval from executive stakeholders.

(2. Phase: Prepare):

Data Source & Privacy: This analysis utilizes reliable, public historical trip data provided by Cyclistic. The extracted datasets cover a 12-month period, from September 2025 to August 2026. To maintain data ethics and ensure rider privacy, all datasets have been strictly stripped of any Personally Identifiable Information (PII) prior to processing.

(Phase 3: Process)

1. Phase Objective To prepare, clean, transform, and validate the consolidated 12-month Cyclistic trip dataset, ensuring high data integrity, eliminating errors and

outliers, and engineering key analytical fields required for the upcoming analysis phase.

2. Tools Used

- **SQL (Google BigQuery):** Used for data consolidation, auditing, cleaning, and feature engineering.
- **Google Docs:** Used for project documentation and process tracking.

3. Data Consolidation

- Merged 12 individual monthly tables (`trip_data_1` through `trip_data_12`) into a unified table named `cyclistic_data.combined_tripdata` using `UNION ALL`.

4. Data Audit & Findings A comprehensive audit was performed on the raw combined dataset prior to finalizing the cleaning steps:

- **Duplicates Check:** Identified 40 duplicate rows using `ride_id` uniqueness verification.
- **Missing Values (NULLs) Check:**
  - Core attributes (`ride_id`, `started_at`, `ended_at`, `member_casual`) contained zero missing values.
  - Missing values were identified in station details (`start_station_name` and `end_station_name`) due to un-docked electric bike trips.
  - **Decision:** Retained these records to avoid losing valid trip duration and user behavioral data.

5. Data Cleaning & Feature Engineering Created the final cleaned table `cyclistic_data.cleaned_tripdata` within a single SQL script implementing:

- **Deduplication:** Applied `ROW_NUMBER() OVER (PARTITION BY ride_id ORDER BY started_at)` filtering for `row_num = 1` to eliminate all 40 duplicate records.
- **Outlier Filtering:** Excluded trip durations <= 0 minutes (system errors or immediate cancellations) and trips > 1,440 minutes (24 hours, representing lost or unreturned bikes).
- **Derived Fields:**
  - `ride_length_minutes`: Calculated total trip length in minutes using `TIMESTAMP_DIFF`.
  - `day_of_week`: Extracted day name (e.g., Sunday, Monday).
  - `month`: Extracted month name (e.g., January, February).

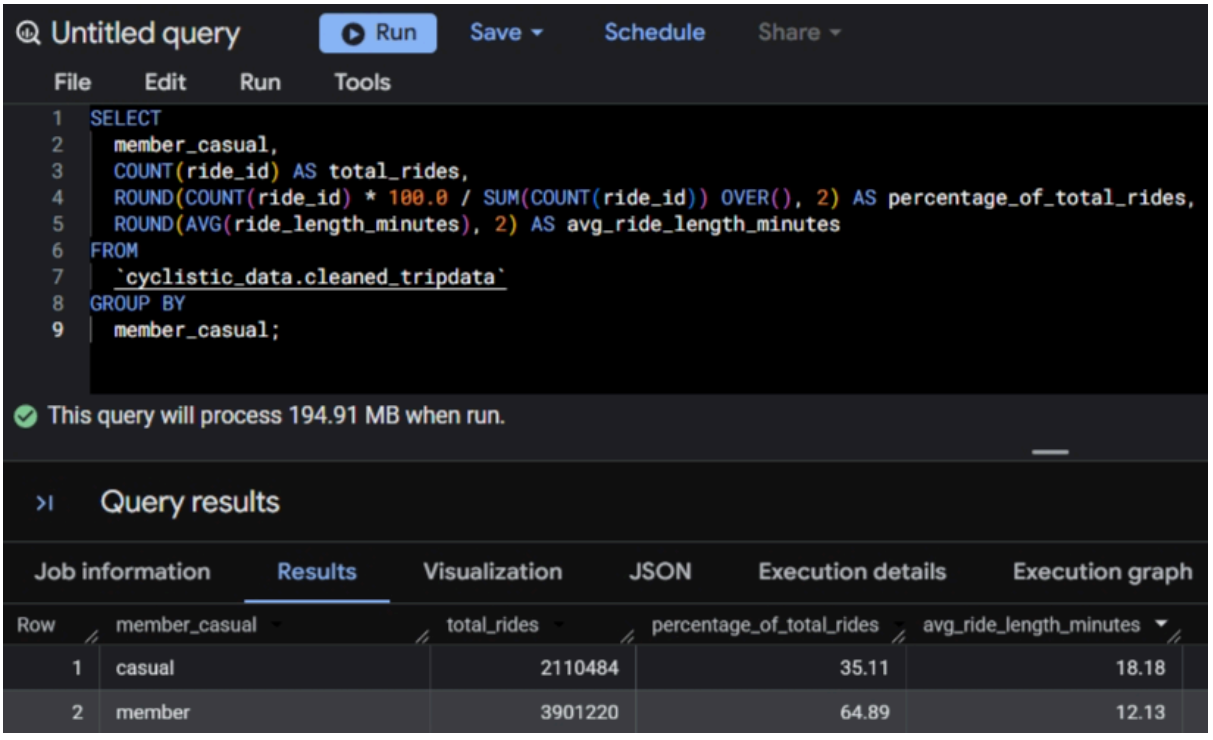
- `year`: Extracted calendar year.

6. Final Validation & Sanity Check

- **Duplicate Count:** Successfully reduced to 0.
- **Trip Duration Range:** Validated to strictly fall between 1 and 1,440 minutes.
- **Dataset Status:** 100% verified, cleaned, and ready for Phase 4 (Analyze).

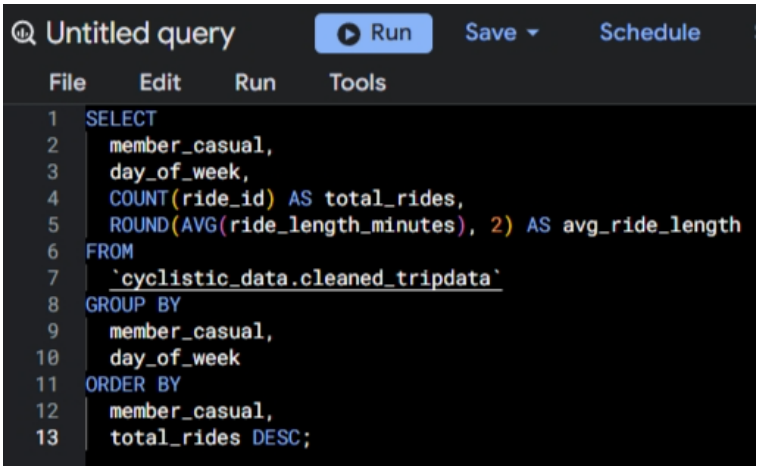
(Phase 4: Analyze Documentation)

1. Phase Objective To analyze the cleaned Cyclistic trip dataset and identify key differences, behavioral patterns, and usage trends between annual members (`member`) and casual riders (`casual`).



1. General Usage Statistics

- **Key Insight:** Annual members generate the majority of total demand (**64.89%** / ~3.9M rides), whereas casual riders account for **35.11%** (~2.1M rides). However, casual riders record significantly longer trip durations, averaging **18.18 minutes** compared to **12.13 minutes** for members.
- **Business Meaning:** Annual members utilize the service for quick, functional point-to-point transportation, while casual riders use it for longer leisure and exploration trips.



Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	member_casual	day_of_week	total_rides	avg_ride_length
1	casual	Saturday	445374	20.55
2	casual	Sunday	339792	20.96
3	casual	Friday	327120	17.83
4	casual	Thursday	262182	15.72
5	casual	Monday	252103	18.35
6	casual	Wednesday	242454	15.12
7	casual	Tuesday	241459	15.9
8	member	Wednesday	623385	11.61
9	member	Tuesday	622449	11.72
10	member	Thursday	616949	11.65
11	member	Friday	570060	12.06
12	member	Monday	558618	11.85
13	member	Saturday	495173	13.38
14	member	Sunday	414586	13.23

2. Day of the Week Analysis

- **Key Insight:** Annual member usage peaks during mid-week working days (Wednesday: **~623K** rides, Tuesday: **~622K** rides) with short, highly consistent trip durations (**~11–12 mins**). Conversely, casual ride volume spikes heavily on weekends (Saturday: **~445K** rides) with trip lengths exceeding **20 minutes**.

- **Business Meaning:** Confirms two distinct user personas—members rely on Cyclistic for routine daily commutes, while casual riders represent weekend recreational and tourism traffic.

Untitled query Run Save Schedule Share

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```sql
SELECT
  member_casual,
  month,
  COUNT(ride_id) AS total_rides,
  ROUND(AVG(ride_length_minutes), 2) AS avg_ride_length
FROM
  `cyclistic_data.cleaned_tripdata`
GROUP BY
  member_casual,
  month
ORDER BY
  member_casual,
  total_rides DESC;
```

This query will process 239.22 MB when run.

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	member_casual	month	total_rides	avg_ride_length
1	casual	July	350059	18.87
2	casual	August	340141	18.17
3	casual	June	301337	19.24
4	casual	September	258623	19.27
5	casual	May	239252	19.47
6	casual	October	218062	17.8

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

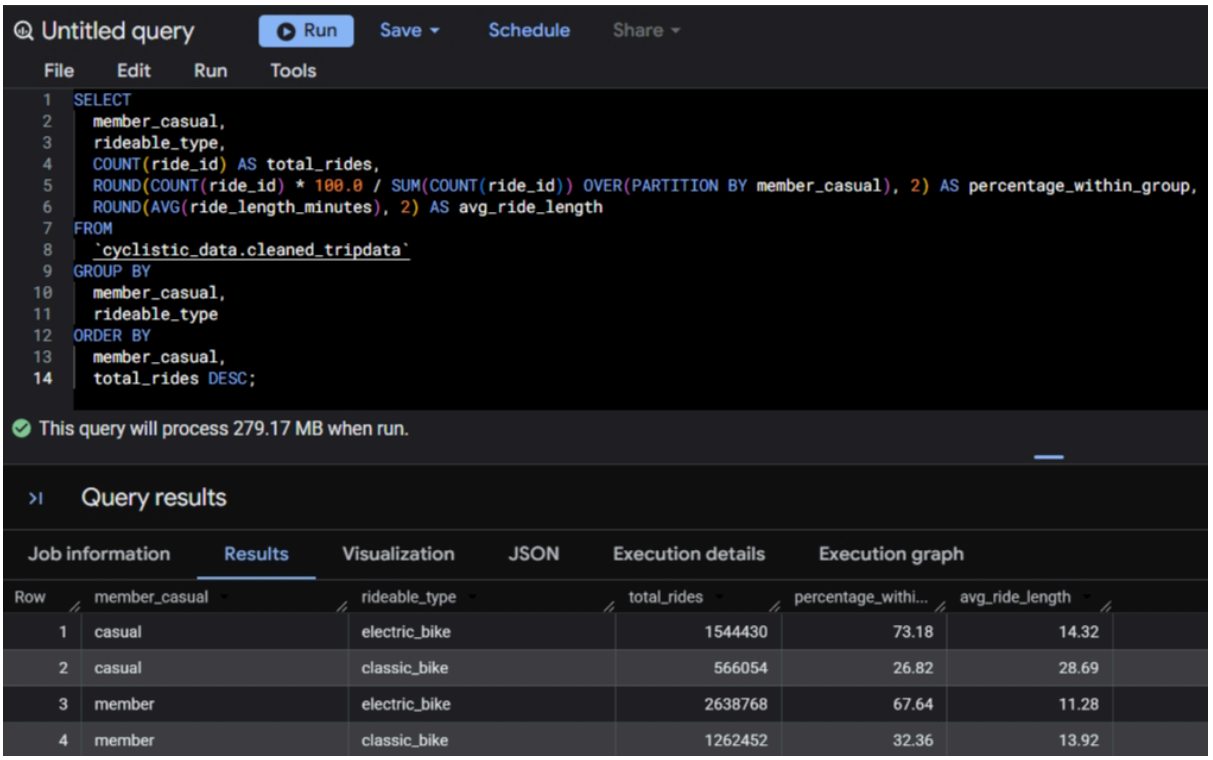
Row	member_casual	month	total_rides	avg_ride_length
7	casual	April	128598	16.49
8	casual	November	96404	15.21
9	casual	March	85943	16.71
10	casual	February	40449	14.04
11	casual	December	27448	12.72
12	casual	January	24168	12.88
13	member	August	514826	12.53
14	member	July	504323	12.52
15	member	June	447529	12.75
16	member	September	443941	12.54
17	member	October	417075	12.04
18	member	May	402653	12.36
19	member	April	311324	11.45
20	member	November	254060	11.59

20	member	November	254060	11.59
21	member	March	225731	11.24
22	member	February	158355	10.69
23	member	January	110971	11.5
24	member	December	110432	11.6

3. Monthly & Seasonal Analysis

- **Key Insight:** Casual ride volume is heavily seasonal, peaking in July (**~350K** rides) during summer holidays and dropping by over **93%** in winter (January: **~24K** rides). Member usage peaks in August (**~514K** rides) upon returning to routines and maintains a resilient baseline throughout winter.

- **Business Meaning:** Casual demand is highly weather and holiday dependent, while member demand is anchored to consistent year-round work and academic schedules.



4. Bike Type Preferences Analysis

- **Key Insight:** Electric bikes are the preferred choice across both segments, accounting for **73.18%** of casual rides and **67.64%** of member rides. Casual riders on classic bikes record the highest overall average duration across the dataset at **28.69 minutes**.
- **Business Meaning:** Highlights strong demand for electric fleet availability, while classic bikes serve extended casual leisure trips.

(Phase 5: Share Documentation)

1. Phase Objective To build an interactive, executive-level dashboard in Tableau that visually communicates key behavioral differences between annual members and casual riders, providing clear visual evidence to support strategic marketing decisions.

2. Tools Used

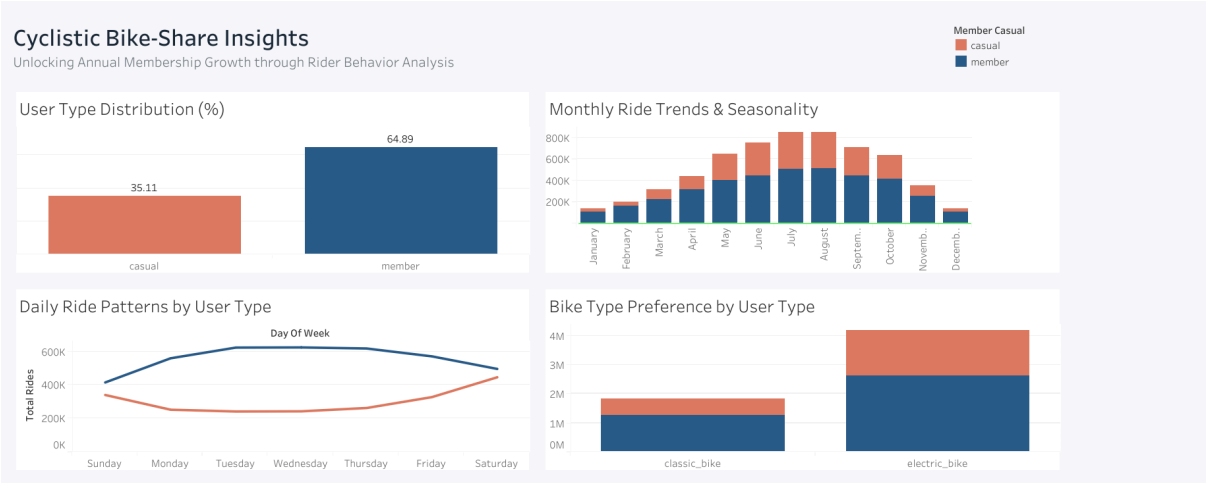
- **Tableau Web Authoring:** Dashboard design, layout formatting, and data visualization.
- **Google Docs:** Comprehensive case study documentation.

3. Dashboard Architecture & Design Elements

- **Executive Grid Layout:** Structured as a balanced 2x2 grid to maximize readability and component alignment without visual clutter.
- **White Card Container Effect:** Applied white sheet backgrounds (`#FFFFFF`) with `8px` outer padding over a subtle light-gray dashboard canvas (`#F8F9FA`) to create high-end visual depth and clear component separation.
- **Distinct Color Palette:** Used Navy Blue (`#2A5C8A`) for Members to represent routine commitment and Coral Orange (`#E07A5F`) for Casuals to symbolize leisure activity.
- **Streamlined Legend:** Floating single legend placed at the top header to eliminate duplicate labels and maximize chart space.

4. Key Visualizations Included

- **User Type Distribution (%):** Highlights overall volume distribution (64.89% Members vs 35.11% Casuals).
- **Daily Ride Patterns by User Type:** Illustrates weekday commuting peaks for Members versus weekend leisure spikes for Casuals.
- **Monthly Ride Trends & Seasonality:** Displays summer peak volume (July/August) and winter drops.
- **Bike Type Preference by User Type:** Demonstrates overwhelming demand for Electric Bikes across both rider segments.



(Phase 6: Act Documentation)

1. Phase Objective To translate analytical findings and visual insights into actionable, data-driven marketing strategies and executive recommendations that achieve the primary business goal of converting casual riders into annual members.

2. Executive Recommendations

- **Targeted Weekend Membership Tier:**
  - **Insight:** Casual usage spikes heavily on weekends with average trip lengths exceeding 20 minutes.
  - **Action:** Introduce a specialized weekend-only annual pass tailored specifically to recreational casual riders, serving as a low-friction entry point to annual subscription models.
- **Seasonality-Driven Conversion Campaigns:**
  - **Insight:** Casual volume peaks in summer (July/August) and begins rising in spring.
  - **Action:** Launch targeted digital marketing campaigns and limited-time conversion discounts during early spring (April/May) to capture casual riders right before peak usage months.
- **Electric Bike Member Perks:**
  - **Insight:** Electric bikes account for 73.18% of casual rides and 67.64% of member rides.
  - **Action:** Provide exclusive member benefits on Electric Bikes—such as priority reservations or discounted per-minute rates—leveraging high electric bike demand to incentivize annual upgrades.
- **In-App Loyalty & Conversion Program:**
  - **Insight:** Casual riders record longer average trip durations (18.18 mins).
  - **Action:** Implement a gamified rewards system in the Cyclistic app where casual riders earn points based on total ride time, redeemable as direct credits toward an annual membership.

3. Next Steps & Implementation

- Present dashboard findings and strategic recommendations to the executive leadership team (Lily Moreno) for budget approval.
- Collaborate with the marketing and product development teams to design campaign creative messaging and configure the new weekend pass in the mobile application.
- Establish quarterly tracking dashboards to monitor casual-to-member conversion rates post-campaign launch.